MACHINE LEARNING

in 30 Days

DAY 1 · VIDEO 1

What is Machine Learning, really?

What is Machine Learning, *really?*

> **In One Line:** Machine Learning is teaching a computer to learn patterns from examples — instead of you writing the rules by hand.

Start here: the chai stall

Picture **Ramesh**, who runs a chai stall outside a Mumbai station.

Nobody gave Ramesh a rulebook. But after 3 years, he *just knows:*

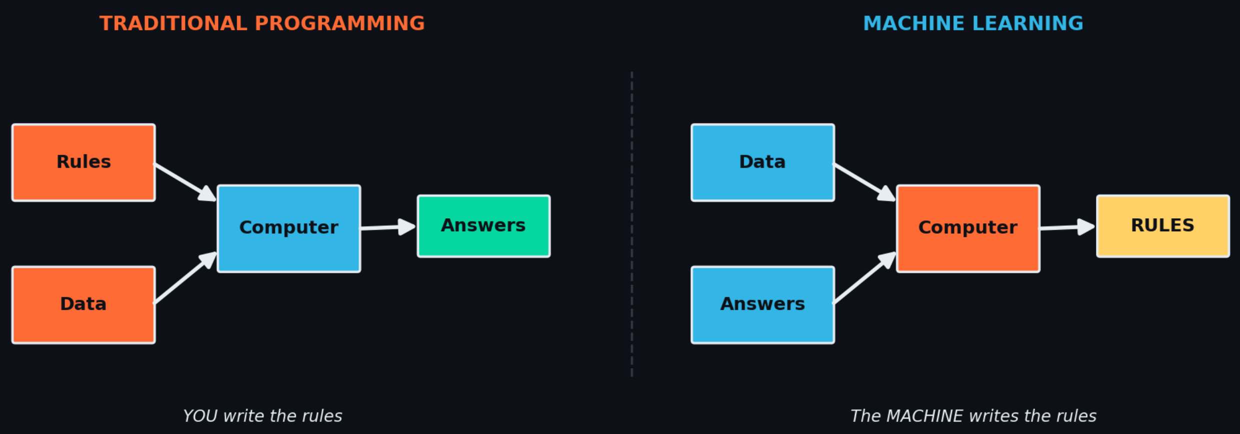
- Rainy evening + train delayed → **sell more chai, keep extra ginger**
- Hot afternoon → **nimbu paani sells, chai doesn't**
- Salary day (1st) → **everyone buys, stock up**

Ramesh was never *programmed*. He **learned from thousands of days of examples.**

> **That is Machine Learning.** A machine, like Ramesh, looking at thousands of past examples and figuring out the pattern **by itself** — so it can predict the future.

The big shift: old way vs ML way

This one picture is the entire reason ML exists:

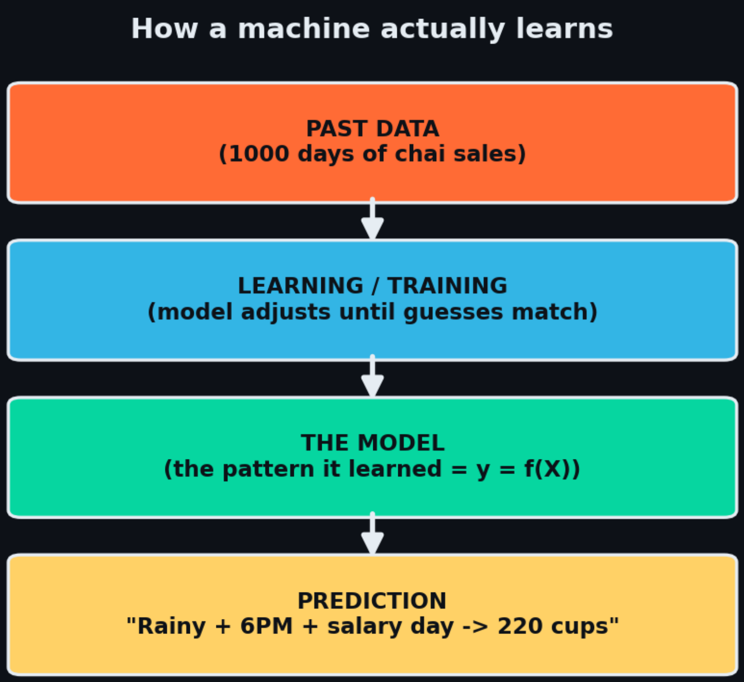


In normal coding YOU write the rules. In ML you give examples and the machine writes the rules itself.

> **The intuition:** In normal coding, *you* are the smart one — you write every rule. In ML, you let the **machine become smart** by showing it examples.

Why this is revolutionary: some problems have too many rules to ever write by hand. How would you write if-else rules to spot a cat in a photo? Impossible. But show a machine 10,000 cat photos and it learns "cat-ness" on its own.

How learning actually happens



Data → Training → Model → Prediction. The "model" is just the pattern it learned.

You already use ML 20 times a day

When you...	ML is quietly working
Open YouTube → perfect next video	Learned your watch history
GPay/PhonePe flags a fraud txn	Learned what fraud looks like
Instagram reels know you too well	Learned your scroll patterns
Gmail auto-sorts spam	Learned from billions of emails

The Math (technical — skip if non-technical, you lose nothing)

Every ML model learns a **function** f that maps input X to output y:

```
        y  =  f(X)

    X = the inputs   (weather, time, crowd)  -> "features"
    y = the answer   (cups of chai sold)      -> "label"
    f = the pattern  (what the machine learns)-> the "model"
```

Training = adjusting f to shrink the gap between its guess and the truth. That gap is the **loss**. All of training is: "make the loss smaller."

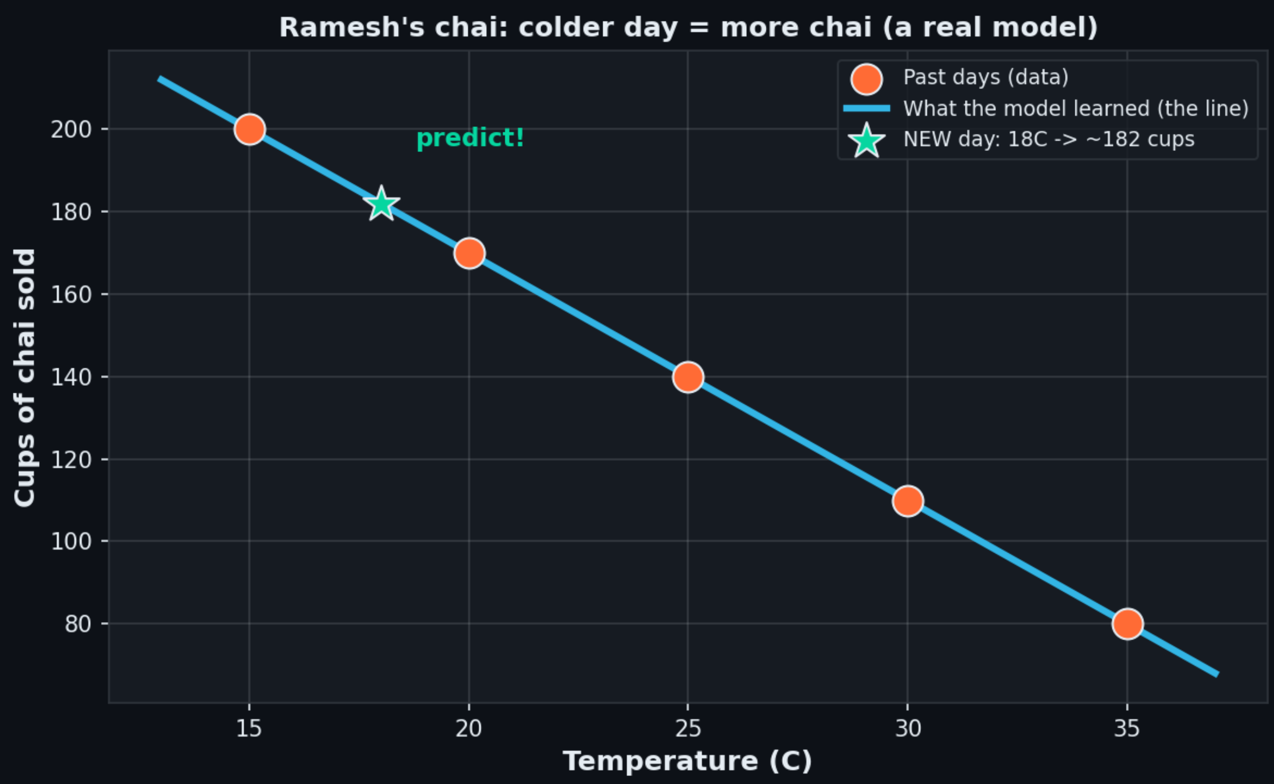
Try it — your first model in 6 lines

```
from sklearn.linear_model import LinearRegression

X = [[15],[20],[25],[30],[35]]   # temperature
y = [200,170,140,110,80]         # cups sold

model = LinearRegression()
model.fit(X, y)                  # <- THIS is learning

print(model.predict([[18]]))     # 18C -> ~182 cups
```



Orange dots = data. Blue line = what the model learned. Green star = a new prediction.

> **model.fit(X, y)** is the entire heart of Machine Learning. Everything in the next 29 days is a more powerful version of it.

Recap — the 20-second version

- ML = learning patterns from examples, not writing rules by hand.
- Old way: you write rules. ML way: the machine finds the rules.
- The learned pattern is called a model ($y = f(X)$).
- You already use it 20× a day — YouTube, GPay, Instagram.
- Training = making wrong guesses less wrong (shrinking the loss).

> Next up — Video 2: The 3 Families of ML: Supervised, Unsupervised, Reinforcement. Just like a child learns 3 ways — see you Day 2.